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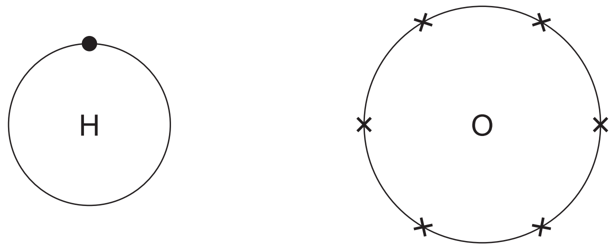
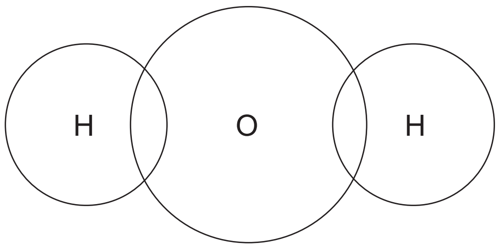
4. *(a)* The table shows the electronic structure of the elements present in water.

Element	Electronic structure
hydrogen	1
oxygen	2,6

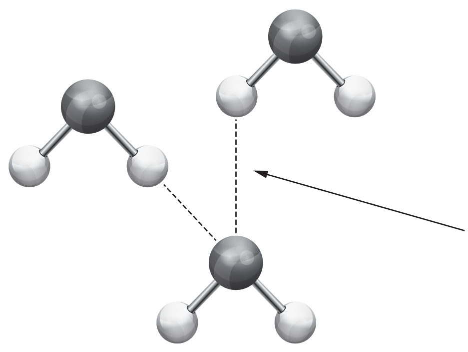
The diagrams show the outer shell electrons in an atom of hydrogen and an atom of oxygen.



(i) Complete the diagram to show the outer shell electrons in a molecule of water. [2]



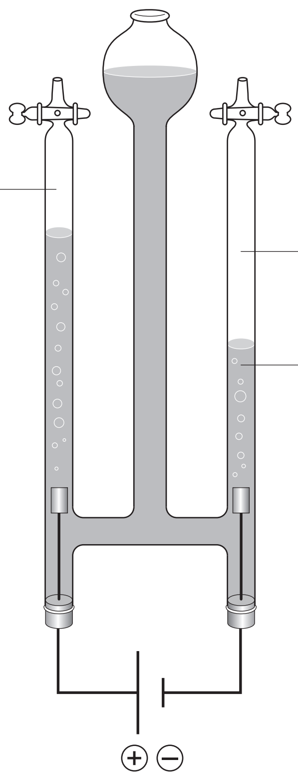
(ii) The diagram shows some water molecules and the weak forces between them.



Put a tick (✓) in the box next to the property of water which can be explained by the weak forces between water molecules. [1]

poor conductor of electricity	
colourless	
good conductor of heat	
low melting point and boiling point	

(b) The diagram shows the apparatus used by a group of students to investigate the volume of hydrogen and oxygen gas formed during the electrolysis of water.



The overall equation for the electrolysis of water is as follows.

$$2H_2O \longrightarrow 2H_2 + O_2$$

(i) The table shows the total volume of hydrogen formed over 10 minutes.

Time (minutes)	0	2	4	6	8	10
Volume of hydrogen (cm^3)	0	10	20	30	40	50

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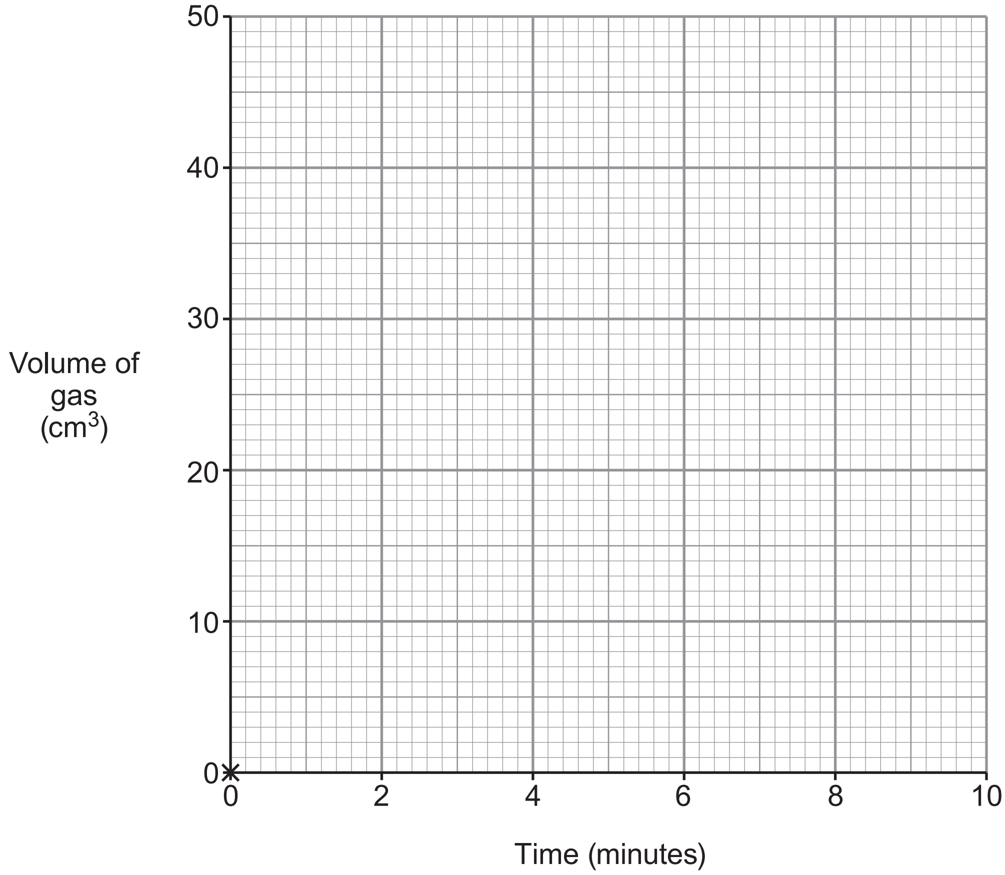
I. Plot the values from the table on the grid. Draw a suitable line. Label this line '**hydrogen**'.

(0,0) has been plotted for you. [2]



II. Using a ruler, draw a second line **on the grid** to show the volume of oxygen that would be collected during the same 10 minutes. Label this line '**oxygen**'. [1]

III. Describe the relationship between the volume of hydrogen and the volume of oxygen formed during electrolysis. [2]

....

....

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3410U201
13

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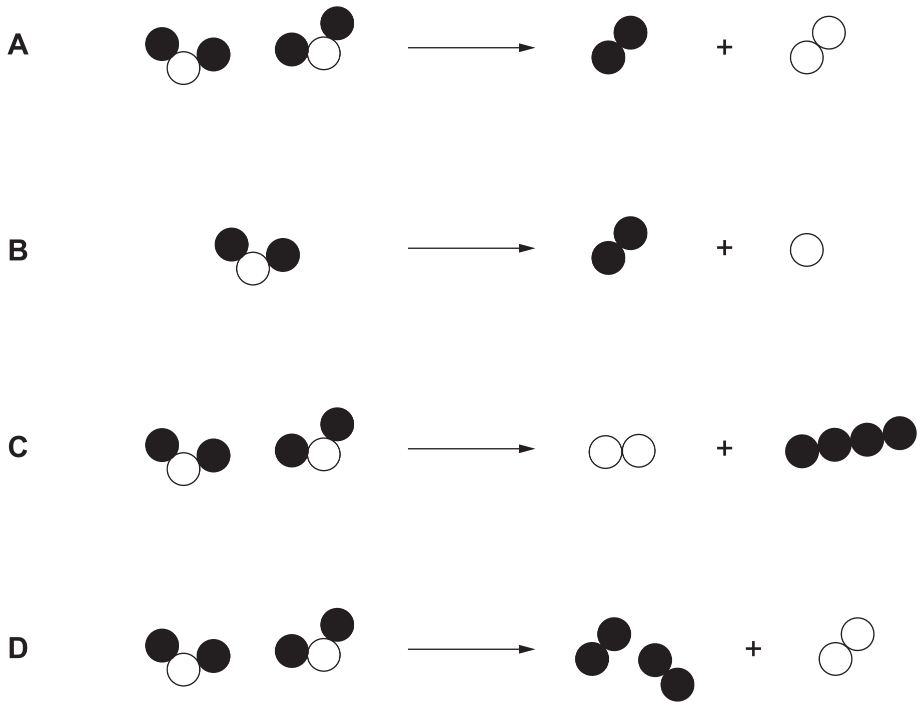
(ii) Electrolysis of water can be represented by the following equation.

$$2H_2O \longrightarrow 2H_2 + O_2$$

Give the **letter**, **A**, **B**, **C** or **D**, of the diagram which also represents this reaction. [1]



Letter ....

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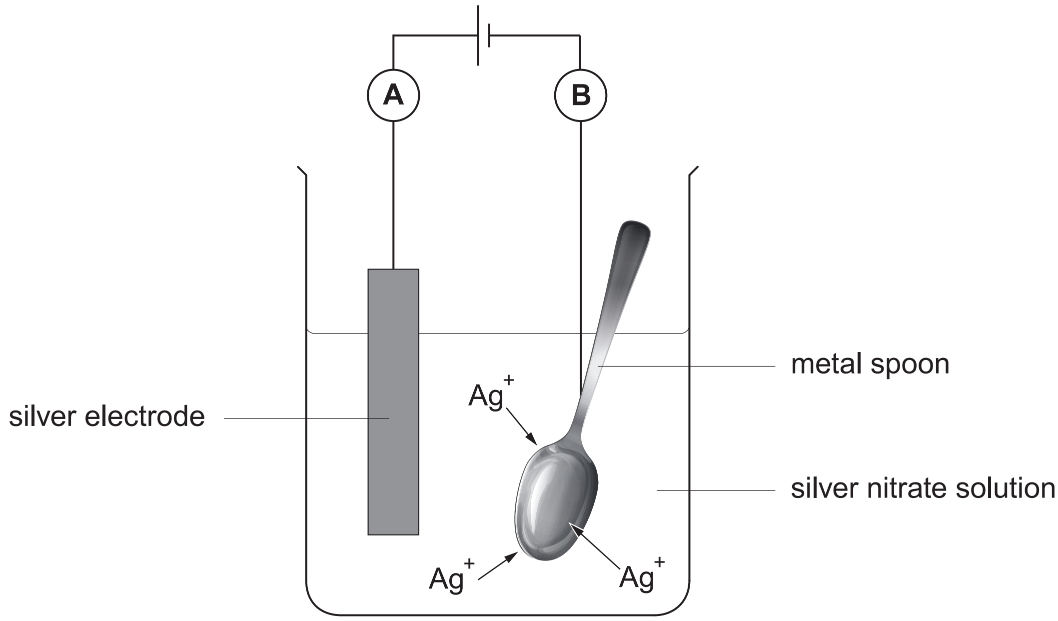
(c) One use of electrolysis is in electroplating. The diagram shows the silver plating of a metal spoon.



3410U201
15

(i) State why this process does not work for a plastic spoon. [1]

....

(ii) Explain why the silver ions move towards the spoon. [2]

....

....

(iii) Put a tick (✓) in the box next to the electrode equation for the reaction at electrode **B**. [1]

$2Ag^{+} + 2e^{-} \longrightarrow Ag$	☐
$Ag^{+} + e^{-} \longrightarrow Ag$	☐
$Ag^{+} - e^{-} \longrightarrow Ag$	☐
$Ag^{+} + 2e^{-} \longrightarrow Ag$	☐

13